

# 34721/34722 Linear Control Design 1

## Spring Semester

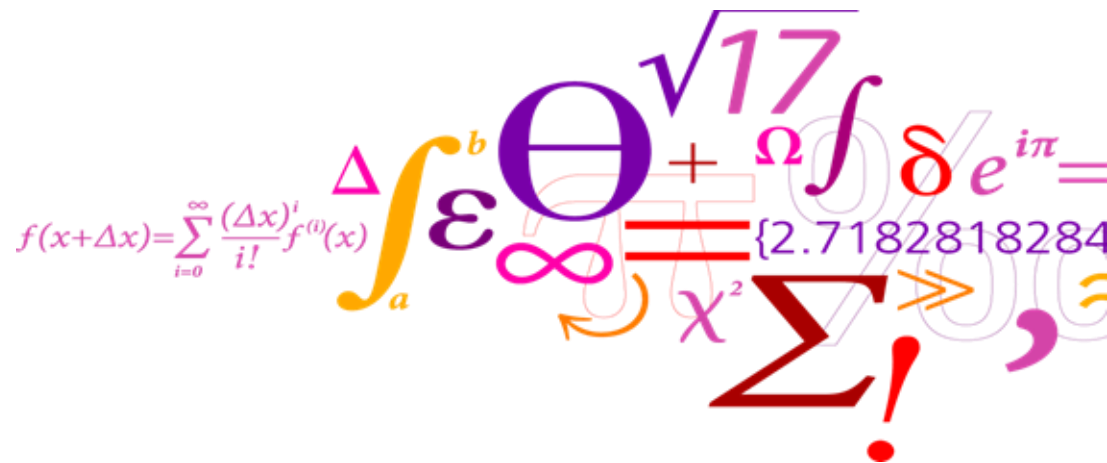
*Silvia Tolu - Dimitrios Papageorgiou*



### Lecture 8/13

**Schedule (13:00-15:00)**

- **PI-Lead design**
- **Exercise**



# What you will learn today

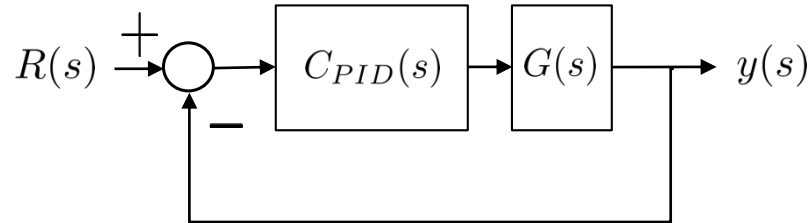
After this lecture you will be able to

- Explain the each part of the PI-Lead controller and their uses.
- Design a P controller based on phase margin specifications.
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- Design a PI-Lead controller based on phase margin specifications.

# PID design – basic concepts

The PID controller has three parts:

- Proportional (main driving force, good for speed of convergence)
- Integral (good for eliminating steady-state error)
- Differential (“almost differential” – good for reducing oscillations) → PI-Lead



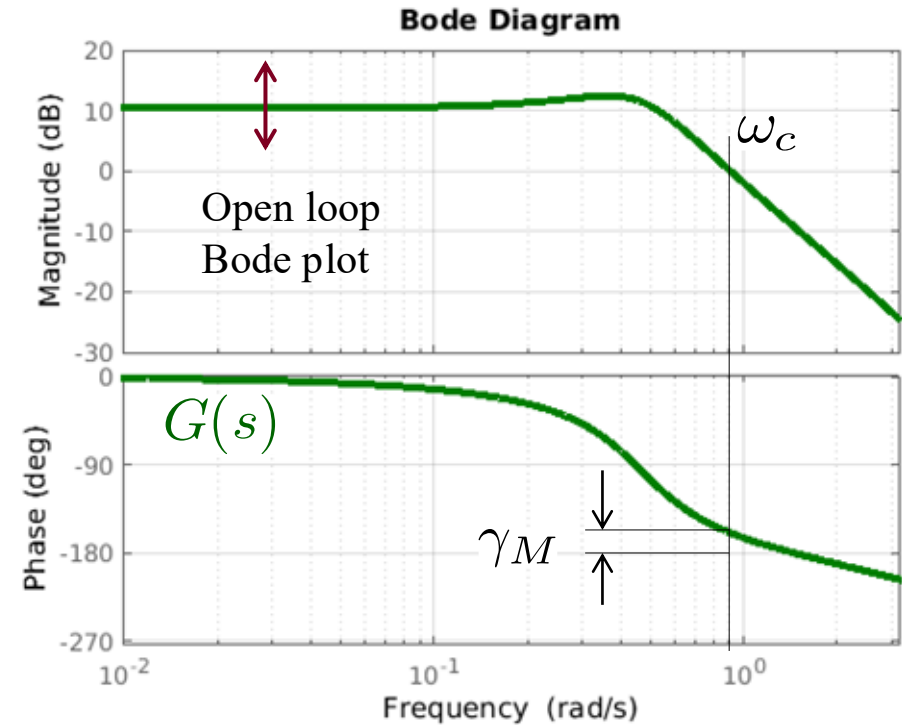
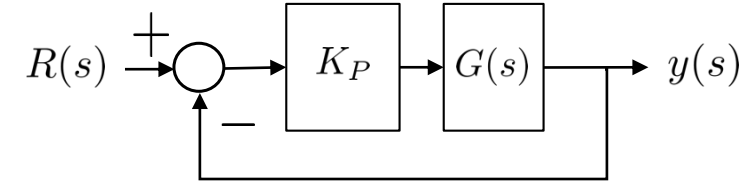
# PI-Lead design – Proportional

## Facts

- The proportional part  $K_P$  can change the magnitude up ( $>1$ ) or down ( $<1$ ) but cannot change the phase (why?).
- It can therefore change the cross-frequency  $\omega_c$  and the phase margin  $\gamma_M$
- Smaller  $K_P$  usually gives better phase and gain margins (the system is more “relaxed”).
- Larger  $K_P$  usually gives smaller static (steady-state) error.

$$G_{ol}(s) = K_P G(s)$$

$$G_{cl}(s) = \frac{K_p G(s)}{1 + K_p G(s)} = \frac{G_{ol}(s)}{1 + G_{ol}(s)}$$



# PI-Lead design – Proportional

## Example

- Consider the system described by the following transfer function

$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}$$

- The open loop transfer function is given by

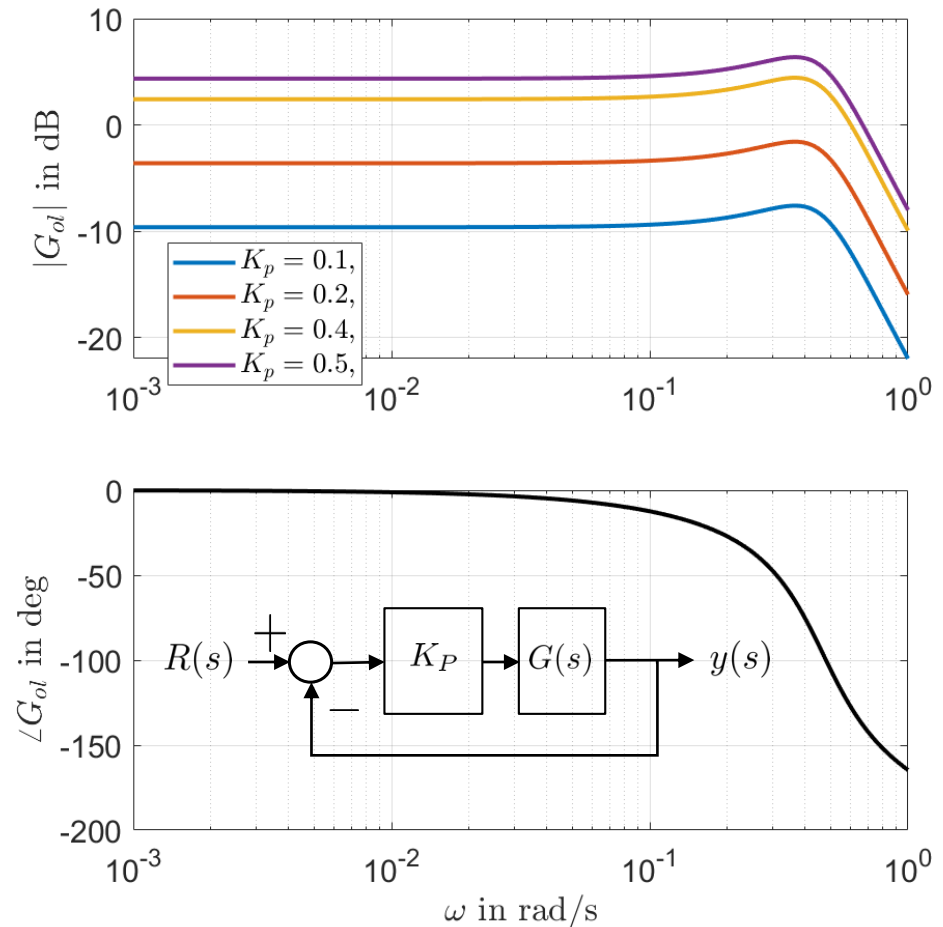
$$G_{ol}(s) = K_P G(s), \quad K_P \in \{0.1, 0.2, 0.4, 0.5\}$$

- The error transfer function is

$$G_e(s) = \frac{1}{1 + G_{ol}} = \frac{1}{1 + K_P G(s)}$$

- The steady-state (static) error for step inputs is

$$e_{ss} = \lim_{s \rightarrow 0} e(s) = \lim_{s \rightarrow 0} s \frac{1}{1 + G_{ol}} \frac{1}{s} = \lim_{s \rightarrow 0} \frac{1}{1 + K_P G(s)}$$



# PI-Lead design – Proportional

## Example

- Consider the system described by the following transfer function

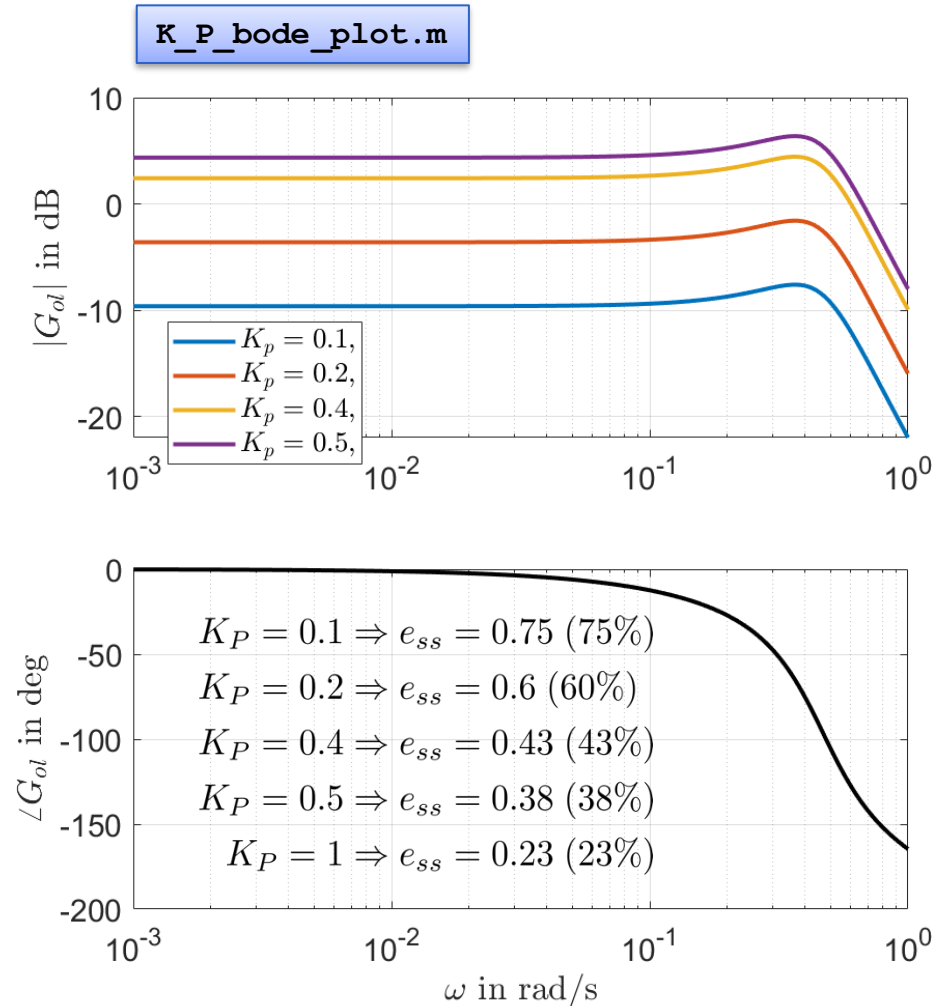
$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}$$

- The open loop transfer function is given by

$$G_{ol}(s) = K_P G(s), \quad K_P \in \{0.1, 0.2, 0.4, 0.5\}$$

- The steady-state (static) error for step inputs is

$$e_{ss} = \lim_{s \rightarrow 0} e(s) = \lim_{s \rightarrow 0} \frac{1}{1 + K_P G(s)} = \frac{1}{1 + K_P 3.3}$$



# PI-Lead design – Proportional

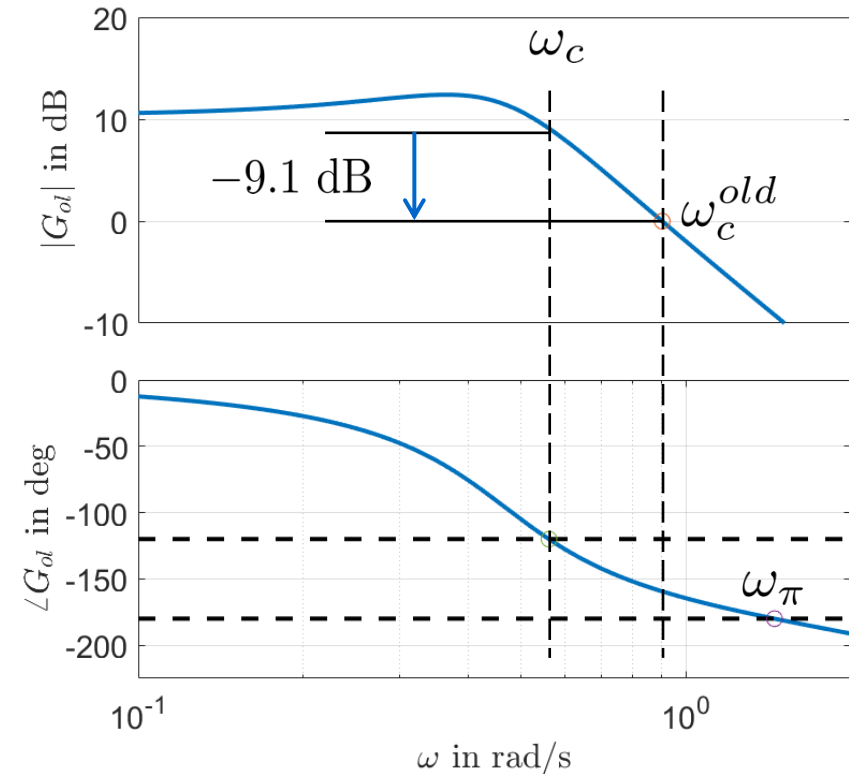
The key point in the design of a P controller is to choose the gain  $K_P$  such that we achieve a desired phase margin.

Example:

$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}$$

- We want  $\gamma_M = 60^\circ$ 
  - Find the crossover frequency at which the phase is  $-180^\circ + 60^\circ = -120^\circ \rightarrow \omega_c = 0.56 \text{ rad/s}$
  - At  $\omega_c$  the magnitude is 9.1 dB. We need to reduce it to 0.
 
$$-9.1 \text{ dB} \rightarrow K_P = 10^{\frac{-9.1}{20}} = 0.3508$$

$$\angle G(j\omega_c) = \gamma_M - \pi \quad \text{Phase balance equation}$$



# PI-Lead design – Proportional

$K_p$  is found to be  $K_P = 0.3508$

$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}$$

- Closed loop transfer function

$$G_{cl}(s) = \frac{0.3508G(s)}{1 + 0.3508G(s)}$$

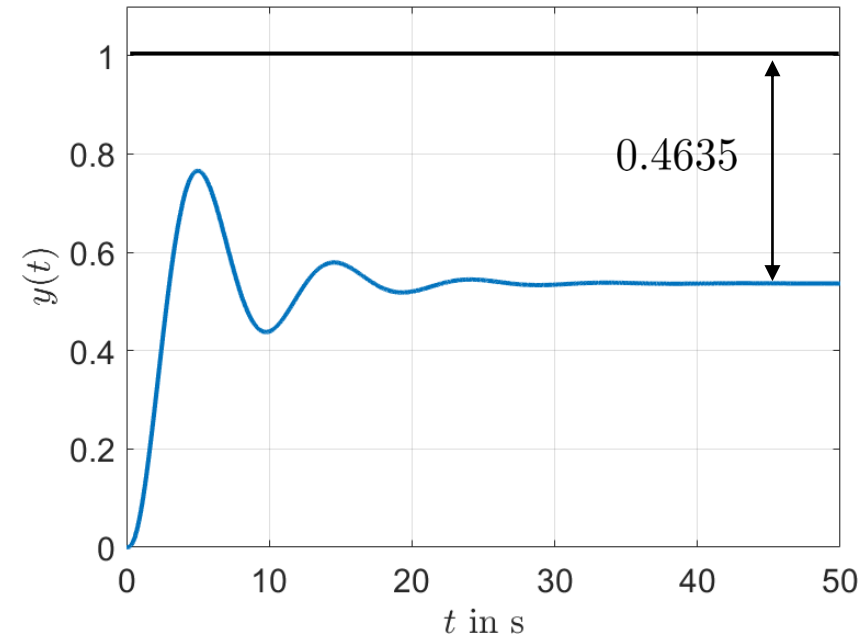
- Error transfer function

$$G_e(s) = \frac{1}{1 + 0.3508G(s)}$$

- Static error

$$e_{ss} = \frac{1}{1 + 0.3508 \cdot 3.3} = 0.4635$$

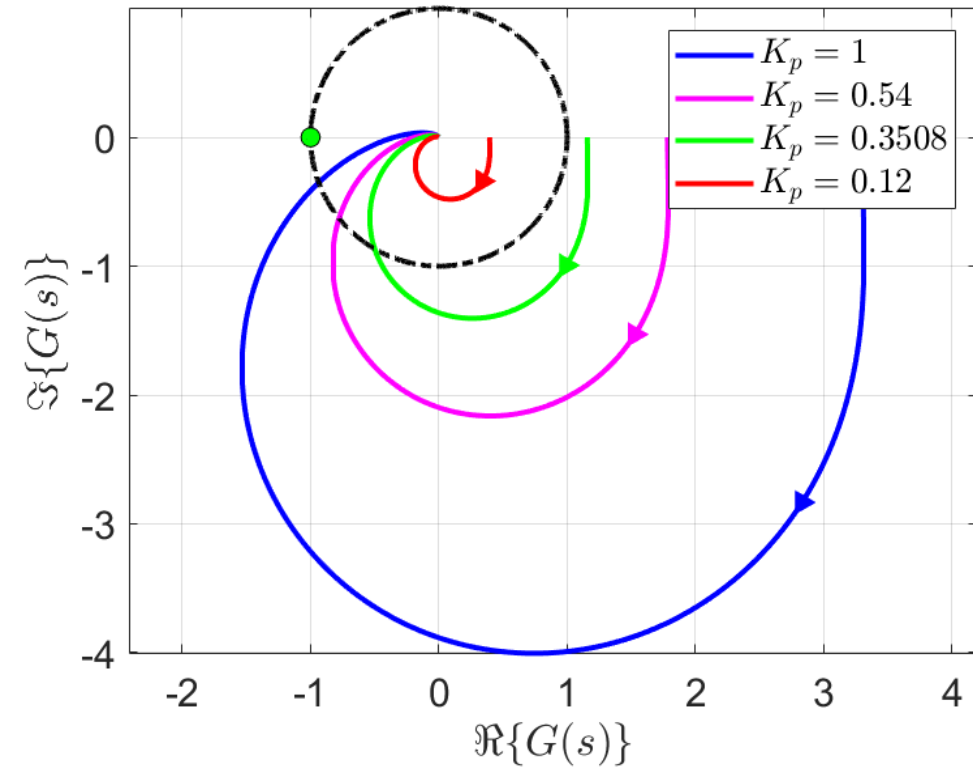
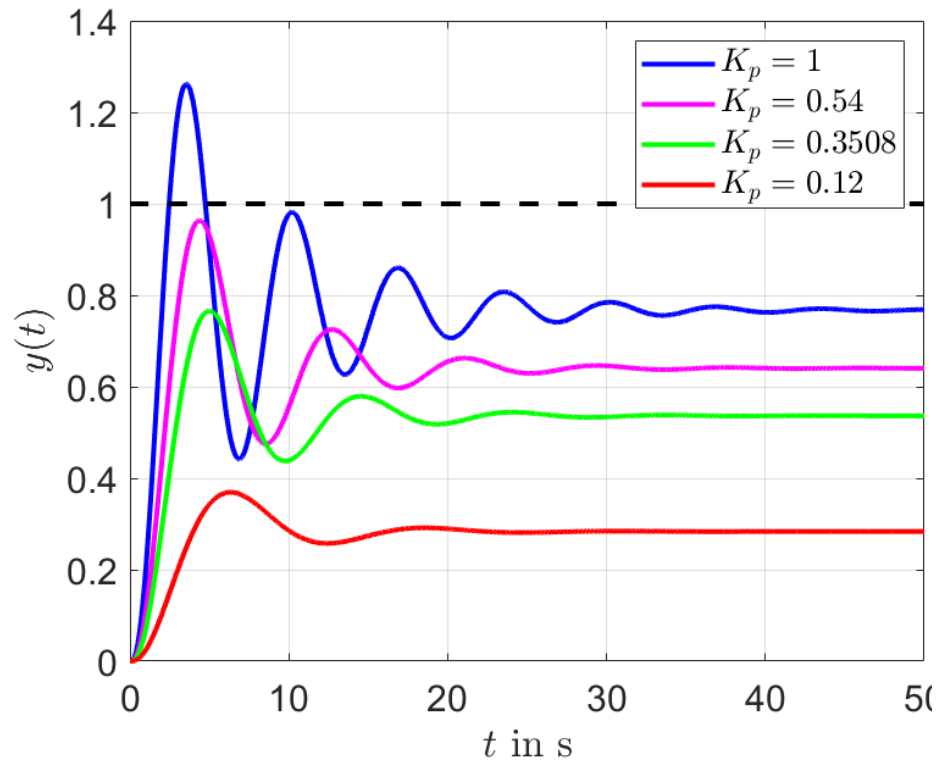
$$G_{DC} = \lim_{s \rightarrow 0} G(s) = 3.3$$



P\_controller\_bode.m

# PI-Lead design – Proportional

$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}, \quad G_{cl}(s) = \frac{K_P G(s)}{1 + K_P G(s)}$$



K\_P\_step\_Nyquist.m

# PI-Lead design – Proportional

Example: Design a P-controller for the system

$$G(s) = \frac{40}{s(s+10)^2}$$

with a phase margin of 60°.

1. What are the poles and zeros of  $G$ ?
2. What should  $K_p$  be?
3. What is the closed-loop transfer function?
4. What is the error transfer function?
5. What is the static (steady-state) error?
6. Plot the step response of the closed-loop system.
7. How much overshoot is there?

Open in Matlab: `Lecture_08_exercise_1.m`

Define  $G(s)$  and answer questions 1-7

## Useful Matlab commands

```
pole()      % finds poles of transfer function
zero()      % finds zeros of transfer function
s = tf('s') % defines s complex variable
tf()        % finds a transfer function
dcgain()    % finds dc gain of transfer function
step()      % plots step response of transfer function
bode()      % plots Bode of transfer function
stepinfo()  % gives info of step response
margin()    % finds gain/phase margins, omega_c, omega_pi
minreal()   % simplifies transfer function
freqresp()  % evaluates transfer function at frequency
```

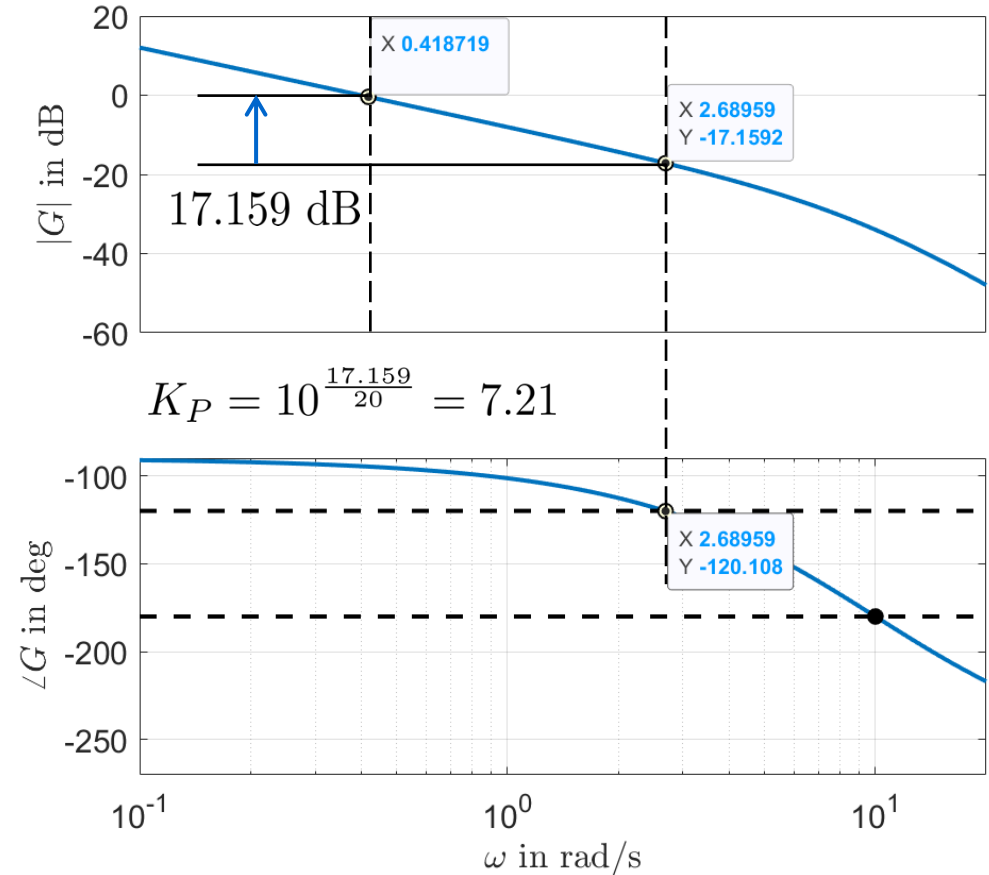
# PI-Lead design – Proportional

Example: Design a P-controller for the system

$$G(s) = \frac{40}{s(s+10)^2}$$

with a phase margin of 60°.

1. What are the poles and zeros of  $G$ ?  $[0, -10, -10]$
2. What should  $K_P$  be?  $K_P = 7.21$
3. What is the closed-loop transfer function?
4. What is the error transfer function?
5. What is the static (steady-state) error?
6. Plot the step response of the closed-loop system.
7. How much overshoot is there?



# PI-Lead design – Proportional

Example: Design a P-controller for the system

$$G(s) = \frac{40}{s(s+10)^2}$$

```
s = tf('s');
G = 40/(s*(s + 10));
```

with a phase margin of  $60^\circ$ .

1. What are the poles and zeros of  $G$ ?  $[0, -10, -10]$
2. What should  $K_P$  be?  $K_P = 7.21$
3. What is the closed-loop transfer function?

$$G_{cl} = K_P G / (1 + K_P G);$$

$$G_{cl}(s) = \frac{K_P G(s)}{1 + K_P G(s)} = \frac{288.4s^3 + 5768s^2 + 2.884 \cdot 10^4 s}{s^6 + 40s^5 + 600s^4 + 4288s^3 + 1.577 \cdot 10^4 s^2 + 2.884 \cdot 10^4 s}$$

4. What is the error transfer function?

$$G_e = 1 / (1 + K_P G);$$

$$G_e(s) = \frac{1}{1 + K_P G(s)} = \frac{s^3 + 20s^2 + 100s}{s^3 + 20s^2 + 100s + 288.4}$$

# PI-Lead design – Proportional

Example: Design a P-controller for the system

Lecture\_08\_exercise\_1\_solution.m

```
s = tf('s');
G = 40/(s*(s + 10)^2);
```

$$G(s) = \frac{40}{s(s + 10)^2}$$

with a phase margin of 60°.

1. What are the poles and zeros of  $G$ ?  $[0, -10, -10]$
2. What should  $K_P$  be?  $K_P = 7.21$
3. What is the closed-loop transfer function?

```
G_cl = K_P*G/(1 + K_P*G);
```

4. What is the error transfer function?

```
G_e = 1/(1 + K_P*G);
```

5. What is the static (steady-state) error?

```
ess = freqresp(s*G_e*1/s,0);
```

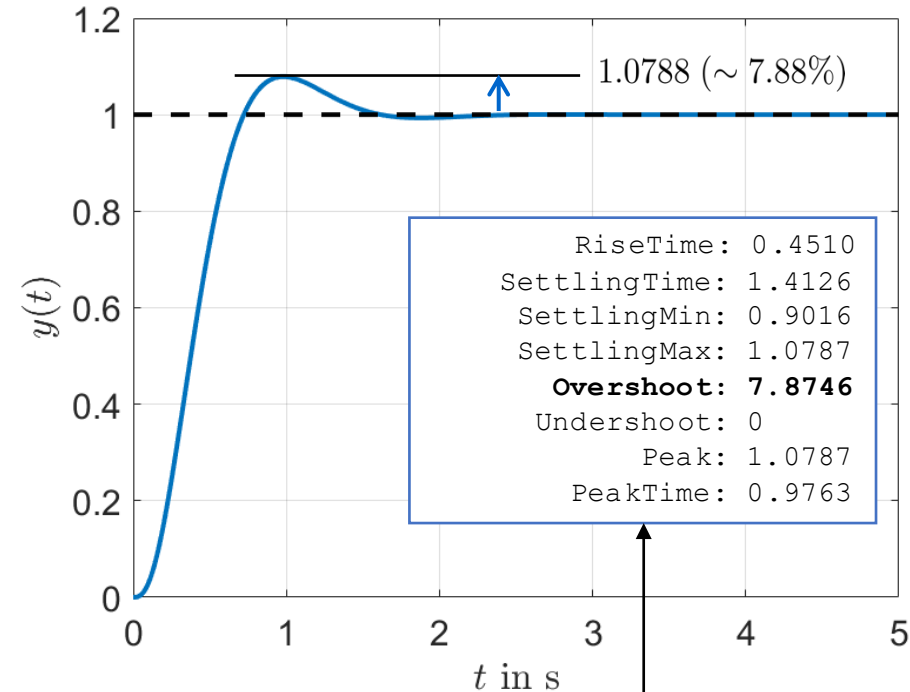
$$e_{ss} = 0$$

6. Plot the step response of the closed-loop system.

```
step(G_cl,0:0.01:5);
```

7. How much overshoot is there?

```
stepinfo(G_cl)
```

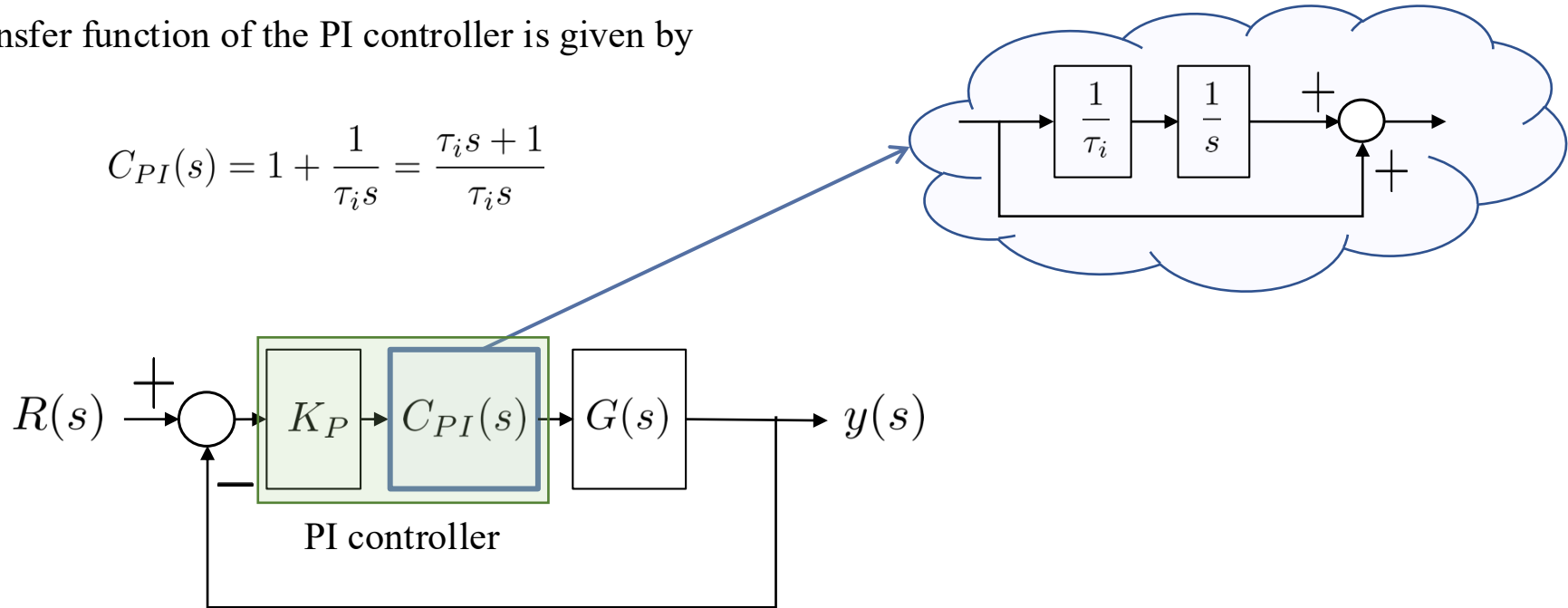


# PI-Lead design – Proportional-Integral

## Facts

- The transfer function of the PI controller is given by

$$C_{PI}(s) = 1 + \frac{1}{\tau_i s} = \frac{\tau_i s + 1}{\tau_i s}$$



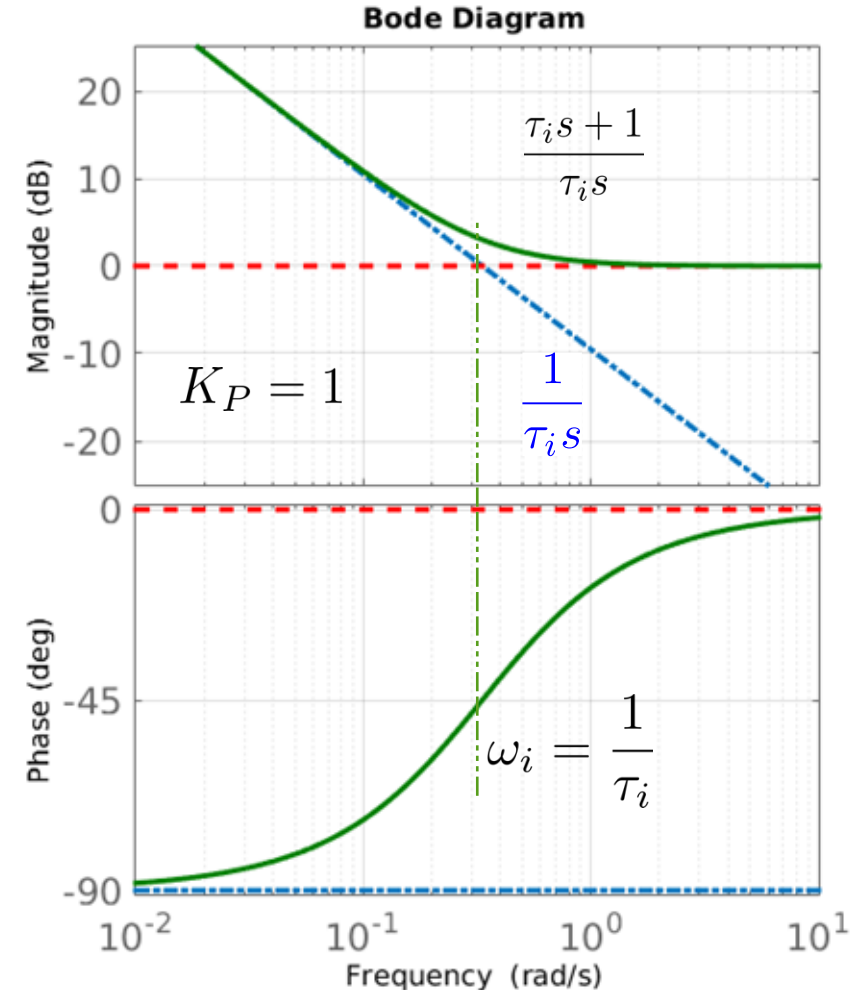
# PI-Lead design – Proportional-Integral

## Facts

- The transfer function of the PI controller (with a gain) is given by
- The integral part  $G_I$  gives large gain at low frequencies ( $\omega_c \rightarrow 0$ ). Small static error. For step inputs  $e_{ss} \rightarrow 0$ .
- The integrator contribution decreases over frequency and the negative phase shift becomes smaller after the break frequency.

$$\omega_i = \frac{1}{\tau_i}$$

- $C_{PI}(s)$  has a pole at 0 and a zero at  $\omega_i$ .



# PI-Lead design – Proportional-Integral

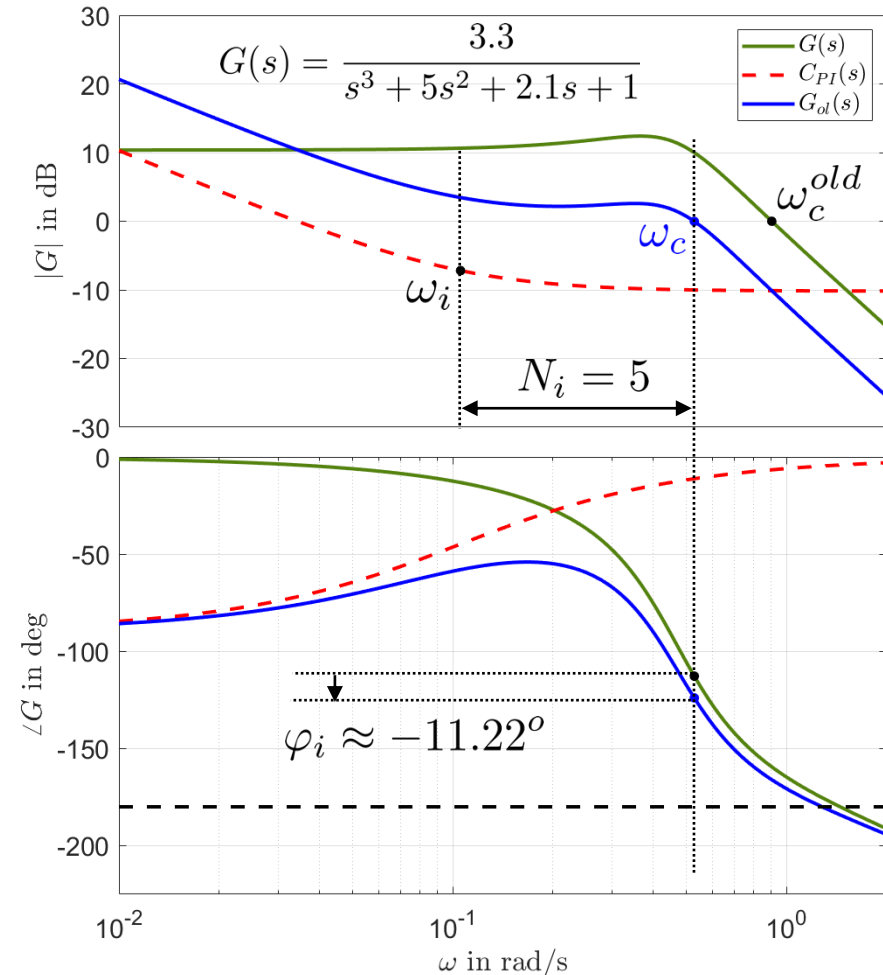
- Given a transfer function  $G$ , the zero of the  $C_{PI}$  at  $\omega_i$  is placed lower than the desired cross-over frequency  $\omega_c$  of  $G$ , by a factor of  $N_i$ , i.e.

$$\omega_i = \frac{1}{\tau_i} = \frac{\omega_c}{N_i} \Leftrightarrow \tau_i = \frac{N_i}{\omega_c}$$

- The phase shift deterioration (**in radians**) at the new cross-over frequency due to the integrator is

$$\angle \left( \frac{1 + j\tau_i\omega_c}{j\tau_i\omega_c} \right) = \angle \left( \frac{1 + jN_i}{jN_i} \right) = \angle \left( 1 - j\frac{1}{N_i} \right)$$

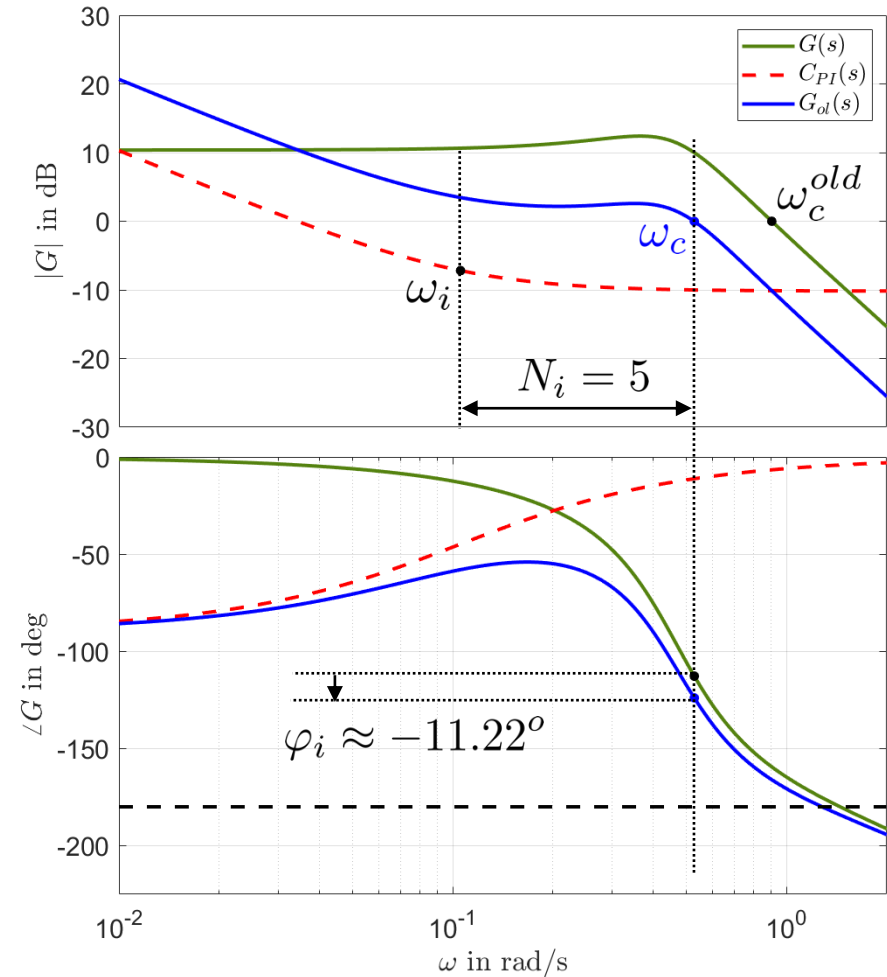
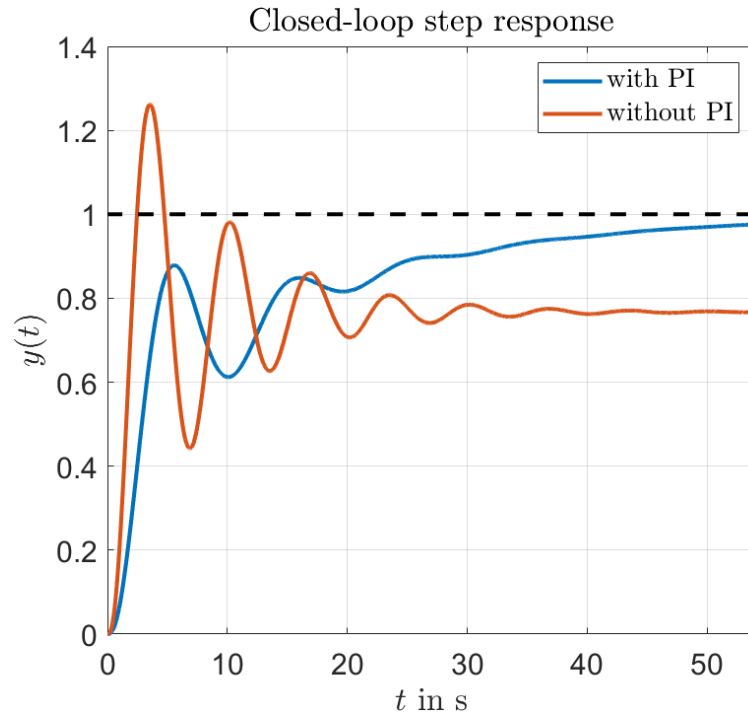
$$\varphi_i = -\arctan \left( \frac{1}{N_i} \right)$$



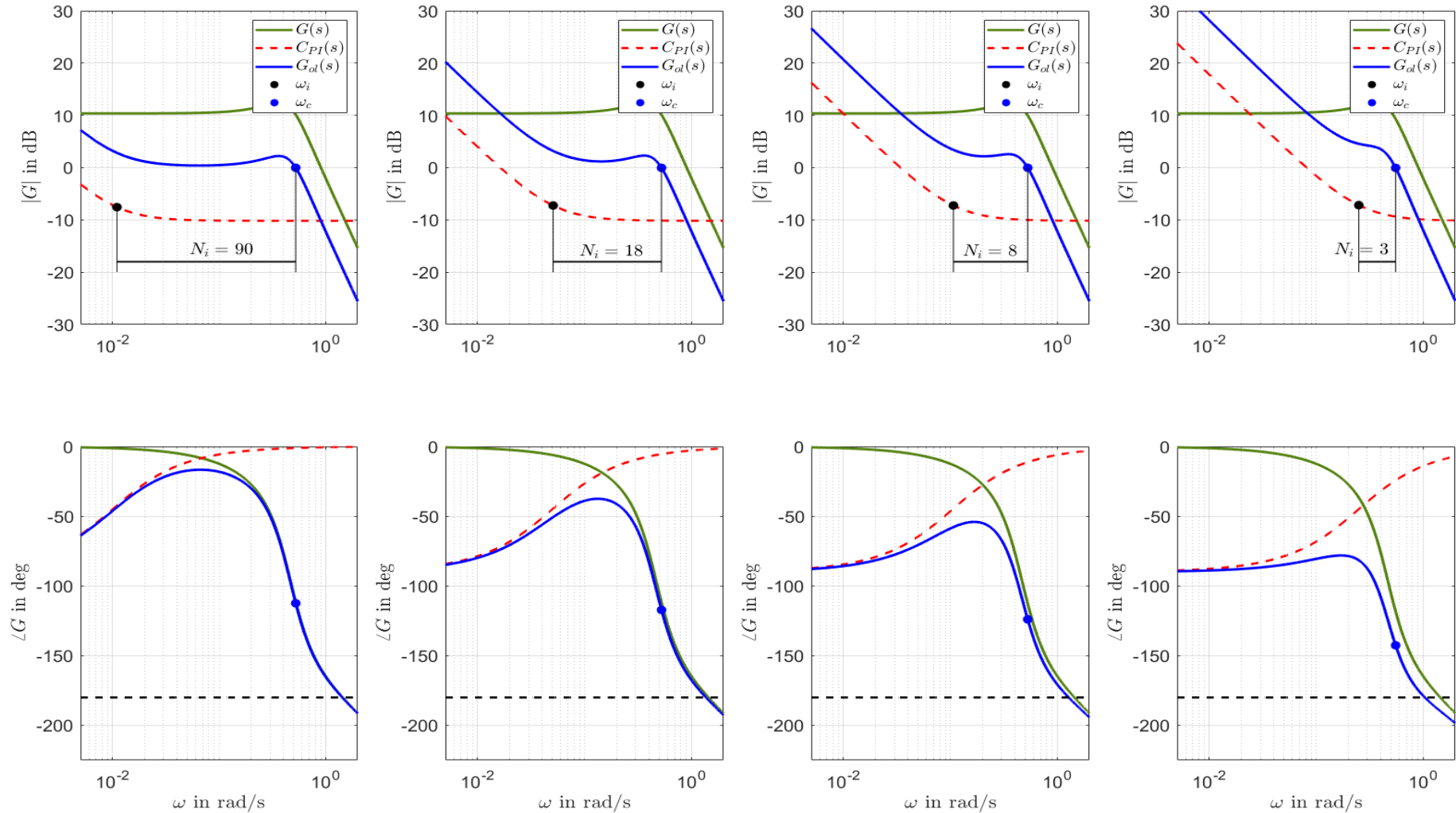
# PI-Lead design – Proportional-Integral

$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}$$

$$C_{PI}(s) = K_P \left( 1 + \frac{1}{9.52s} \right), \quad K_P = 0.31$$



# PI-Lead design – Proportional-Integral



# PI-Lead design – Proportional-Integral

- For a given  $N_i$ , we can find the PI phase contribution.
- For a desired phase margin  $\gamma_M$  the phase of the **(without the  $K_P$ )** open-loop transfer function  $G(s)C_{PI}(s)$  at the new crossover frequency  $\omega_c$  will be

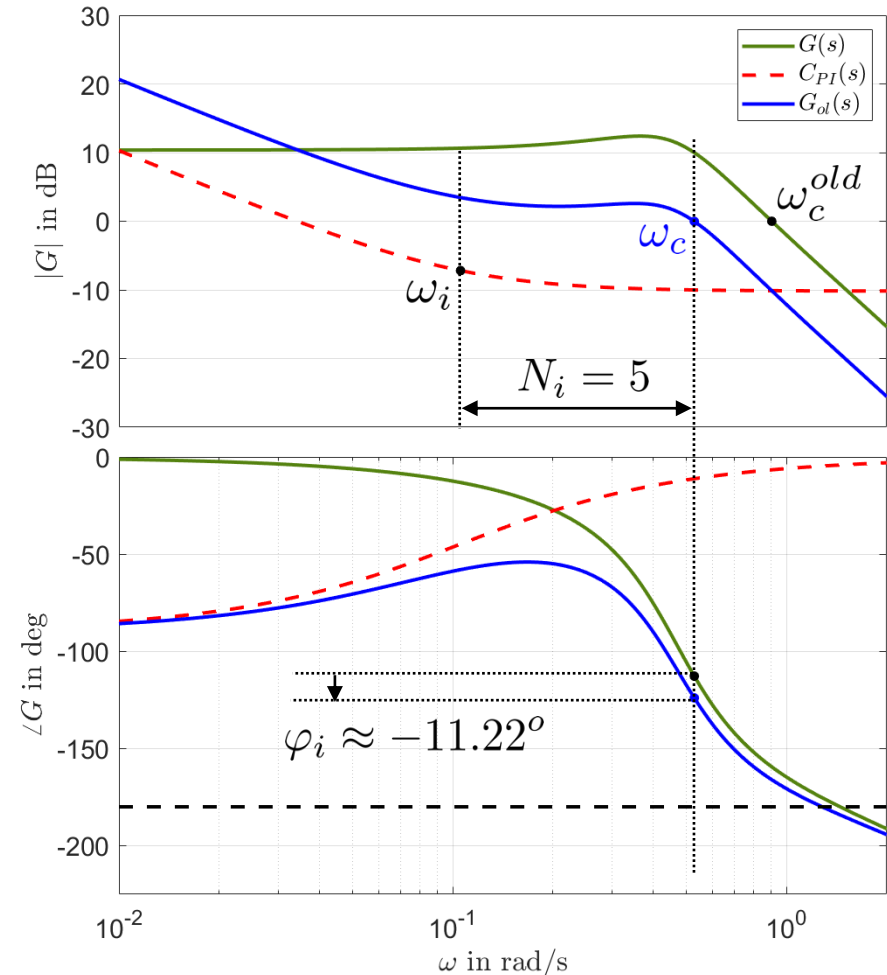
$$\begin{aligned}\angle G_{ol}(j\omega_c) &= \angle G(j\omega_c) + \angle C_{PI}(j\omega_c) \Leftrightarrow \\ -\pi + \gamma_M &= \angle G(j\omega_c) - \tan^{-1}\left(\frac{1}{N_i}\right) \Leftrightarrow\end{aligned}$$

**Phase balance equation**

$$\angle G(j\omega_c) = \gamma_M - \pi + \tan^{-1}\left(\frac{1}{N_i}\right)$$

- The new crossover frequency  $\omega_c$  can be calculated from the bode plot of  $G(s)$ .
- Once  $\omega_c$  is known, we can find  $\tau_i, K_P$  from

$$\tau_i = N_i \frac{1}{\omega_c}, \quad K_P = \frac{1}{|C_{PI}(s)G(s)|} \Big|_{s=j\omega_c}$$



# PI-Lead design – Proportional-Integral

Example: Design a PI controller for the system

$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}$$

with  $N_i = 3$  if we want a phase margin of  $60^\circ$ .

1. What are the poles and zeros of  $G$ ?
2. Find the phase contribution  $\varphi_i$  of the PI.
3. Find the new crossover frequency  $\omega_c$ .
  - **Tip:** *Bode plot may be useful*
4. Find the PI time constant  $\tau_i$ .
5. Find the proportional gain  $K_P$ .
  - **Tip:** *Find the PI transfer function*
  - **Tip:** *Remember that  $G_{ol}(\omega_c) = 0$  dB*

Open in Matlab: `Lecture_08_exercise_2.m`

Define  $G(s)$  and answer questions 1-5

## Useful Matlab commands

```
pole()      % finds poles of transfer function
zero()      % finds zeros of transfer function
s = tf('s') % defines s complex variable
tf()        % finds a transfer function
dcgain()    % finds dc gain of transfer function
step()      % plots step response of transfer function
bode()      % plots Bode of transfer function
stepinfo()  % gives info of step response
margin()    % finds gain/phase margins, omega_c, omega_pi
minreal()   % simplifies transfer function
freqresp()  % evaluates transfer function at frequency
```

# PI-Lead design – Proportional-Integral

Example: Design a PI controller for the system

$$G(s) = \frac{3.3}{s^3 + 5s^2 + 2.1s + 1}$$

with  $N_i = 3$  if we want a phase margin of  $60^\circ$ .

1. What are the poles and zeros of  $G$ ?

$$[-4.5899, -0.205 \pm j0.4193]$$

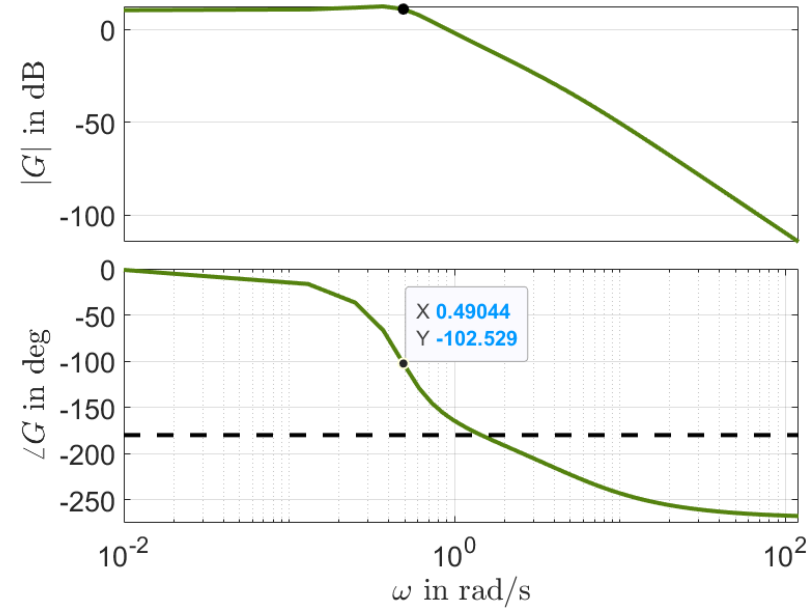
2. Find the phase contribution  $\varphi_i$  of the PI.

$$\varphi_i = -\arctan\left(\frac{1}{N_i}\right) = -\arctan\left(\frac{1}{3}\right) = -0.3218 \text{ rad} = -18.4349^\circ$$

3. Find the new crossover frequency  $\omega_c$ .

$$\angle G(j\omega_c) = \gamma_M - \varphi_i - 180^\circ = 60^\circ + 18.4349^\circ - 180^\circ = -101.5651^\circ \Rightarrow \omega_c = 0.49044 \text{ rad/s}$$

4. Find the PI time constant  $\tau_i$ .  $\tau_i = N_i \frac{1}{\omega_c} = 6.117$



Lecture\_08\_exercise\_2\_solution.m

5. Find the proportional gain  $K_P$ .

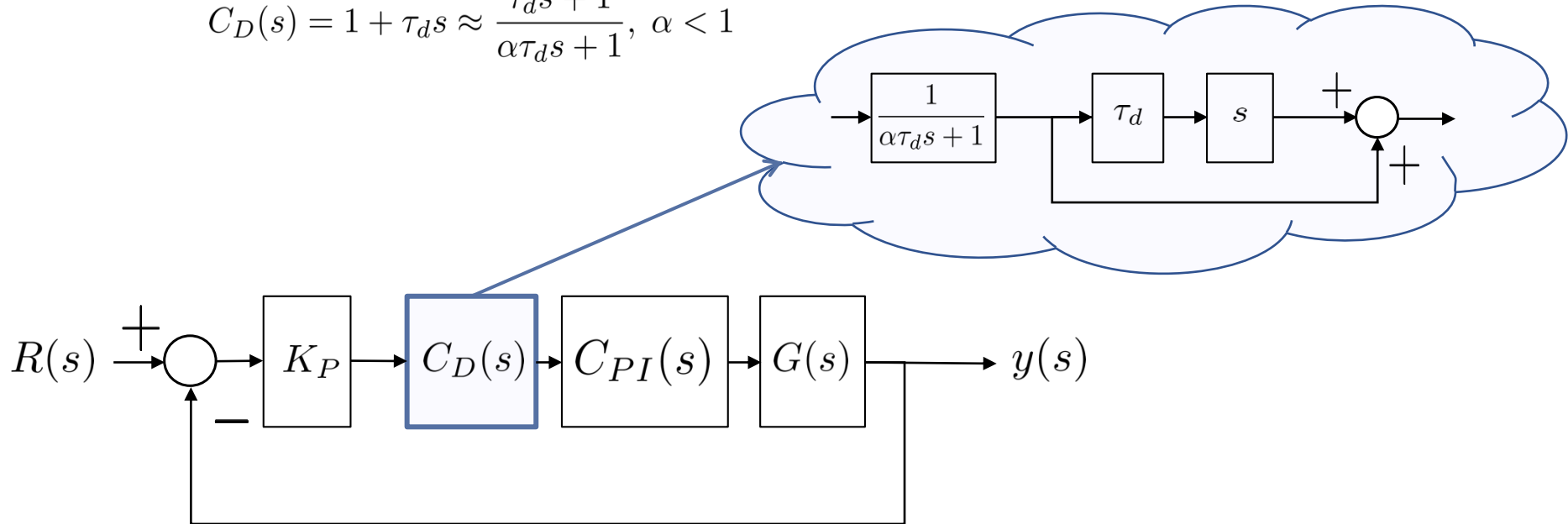
$$K_P = \frac{1}{|C_{PI}(s)G(s)|} \Big|_{s=j\omega_c} = 0.2686$$

# PI-Lead design – Proportional-Integral-“Differential”

## Facts

- The transfer function of the D-part of the PID (PI-Lead) controller is given by

$$C_D(s) = 1 + \tau_d s \approx \frac{\tau_d s + 1}{\alpha \tau_d s + 1}, \alpha < 1$$



# PI-Lead design – Proportional-Integral-“Differential”

## Facts

- The transfer function of the D-part of the PID (PI-Lead) controller is given by

$$C_D(s) = \underbrace{1 + \tau_d s}_{\text{“PD”}} \approx \underbrace{\frac{\tau_d s + 1}{\alpha \tau_d s + 1}}_{\text{“P-Lead”}}, \quad \alpha < 1$$

- A Lead controller has a zero at  $\omega_d = \frac{1}{\tau_d}$  and a pole at  $\omega_a = \frac{1}{\alpha \tau_d}$  (higher than the zero).
- The zero give positive phase shift (improves the phase margin).
- The pole is added (hence we change from “D” to “Lead”) such that we avoid very high gains at high frequency (  $|C_D(\omega)| \rightarrow \infty$  as  $\omega \rightarrow \infty$ ):
  - The controller is not implementable
  - Amplification of noise

# PI-Lead design – Proportional-Integral-“Differential”

## Facts

- The transfer function of the D-part of the PID (PI-Lead) controller is given by

$$C_D(s) = \underbrace{1 + \tau_d s}_{\text{“PD”}} \approx \underbrace{\frac{\tau_d s + 1}{\alpha \tau_d s + 1}}_{\text{“Lead”}}, \quad \alpha < 1$$

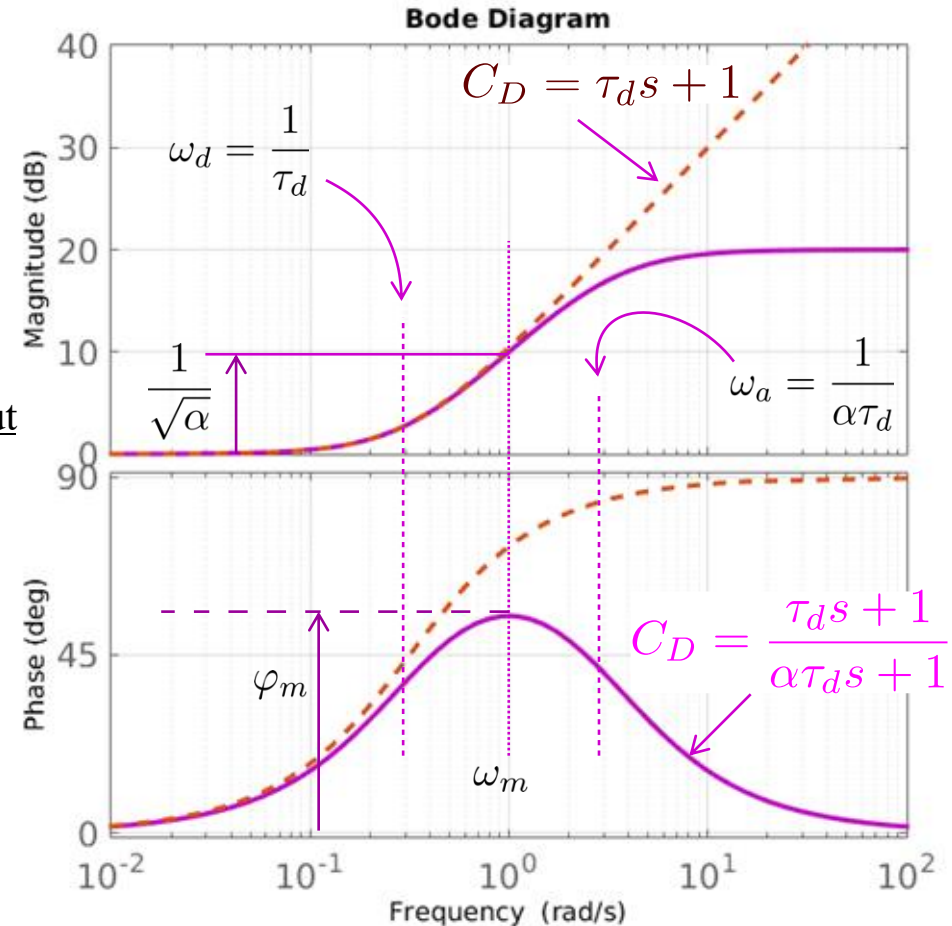
- With the Lead part we achieve improvement of the phase margin (less overshoot and oscillations) without amplification of noise and high frequency content.

- Maximum positive phase shift at  $\omega_m = \frac{1}{\tau_d \sqrt{\alpha}}$

$$\varphi_m = \arctan\left(\frac{1 - \alpha}{2\sqrt{\alpha}}\right) = \arcsin\left(\frac{1 - \alpha}{1 + \alpha}\right)$$

- Amplitude contribution (not in dB) at  $\omega_m$  is

$$|C_D(j\omega_m)| = \frac{1}{\sqrt{\alpha}}$$



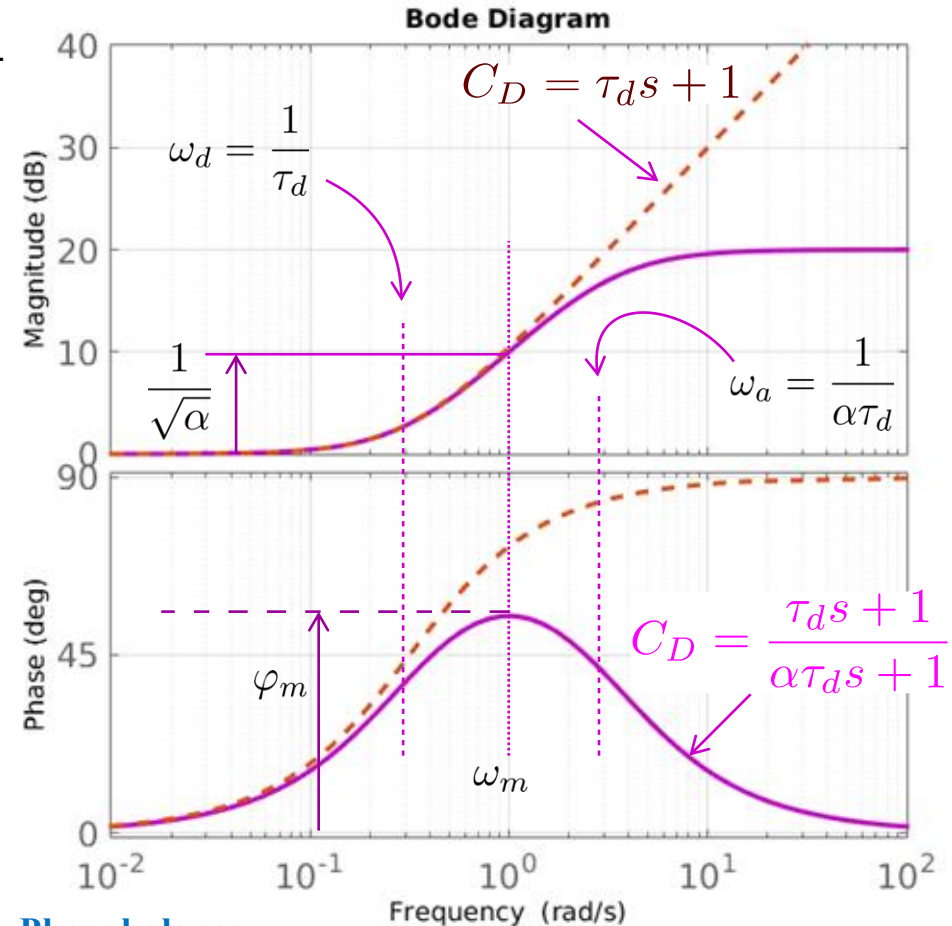
# PI-Lead design – Proportional-Integral-“Differential”

- The positive phase shift from Lead part should be exploited at the new crossover frequency of the open-loop transfer function (why?)
- For a given  $N_i$  and  $\alpha$ , we can find the phase contribution from the PI and Lead parts.
- For a desired phase margin  $\gamma_M$  the phase of the (**without the  $K_p$** ) open-loop transfer function  $G(s)C_{PI}(s)C_D(s)$  at the new cross-over frequency  $\omega_c$  will be

$$\angle G_{ol}(j\omega_c) = \angle G(j\omega_c) + \angle C_{PI}(j\omega_c) + \angle C_D(j\omega_c) \Leftrightarrow$$

$$-\pi + \gamma_M = \angle G(j\omega_c) - \tan^{-1}\left(\frac{1}{N_i}\right) + \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) \Leftrightarrow$$

$$\angle G(j\omega_c) = \gamma_M - \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) - \pi + \tan^{-1}\left(\frac{1}{N_i}\right)$$



Phase balance  
equation

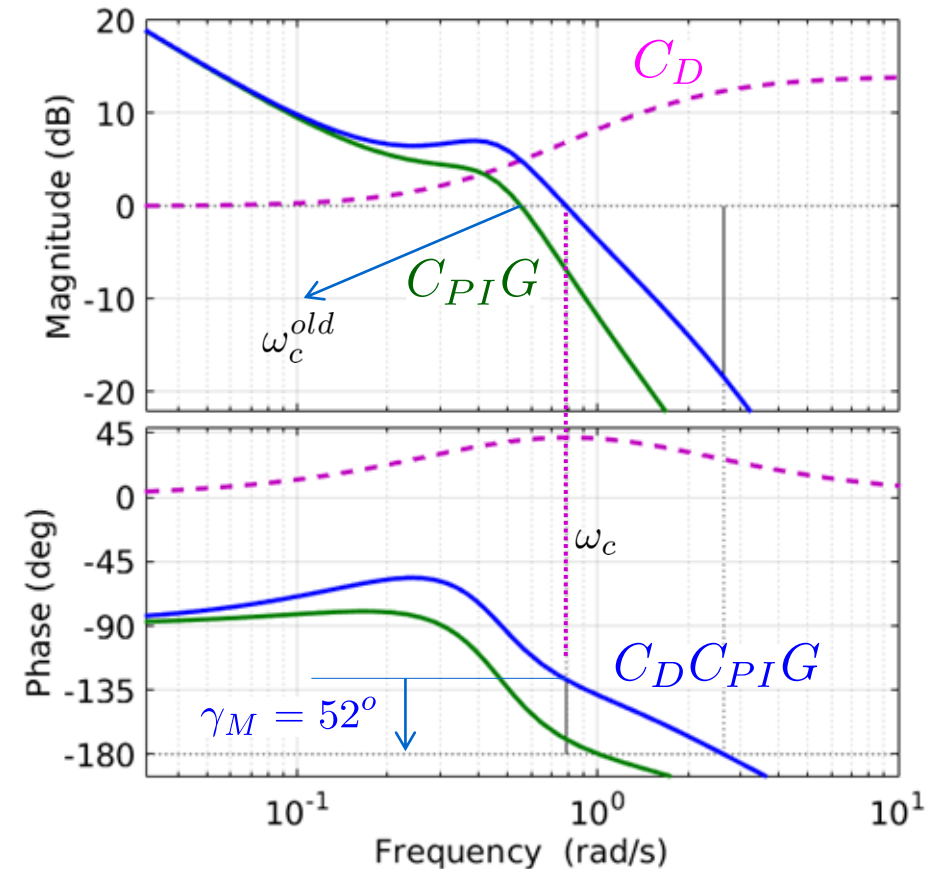
# PI-Lead design – Proportional-Integral-“Differential”

- The new crossover frequency  $\omega_c$  can be calculated from the bode plot of  $G(s)$ .
- Once  $\omega_c$  is known, we can find  $\tau_i, \tau_d, K_P$  from

$$\tau_d = \frac{1}{\omega_c \sqrt{\alpha}}$$

$$\tau_i = N_i \frac{1}{\omega_c}$$

$$K_P = \frac{1}{|C_{PI}(s)C_D(s)G(s)|} \Big|_{s=j\omega_c}$$



# PI-Lead design – Proportional-Integral-“Differential”

Example: Design a PI-Lead controller for the system

$$G(s) = \frac{40}{(s+1)(s+10)^2}$$

Open in Matlab: `Lecture_08_exercise_3.m`

Define  $G(s)$  and answer questions 1-6

where the following (typical) choices are made:

$$\gamma_M = 60^\circ, N_i = 3, \alpha = 0.3$$

1. What is the phase shift from the PI-part at the crossover frequency?
2. What is the maximum phase shift from the P-Lead-part?
3. What is the phase of  $G(s)$  at the frequency where we want to have the given phase margin (i.e. at the new crossover frequency of the open-loop transfer function)?
4. What is the new crossover frequency of the open-loop transfer function?
5. What should the PI-Lead parameters  $\tau_i, \tau_d, K_P$  be for this control objective?
6. Plot the response of the closed-loop system to a step input.

# PI-Lead design – Proportional-Integral-“Differential”

Example: Design a PI-Lead controller for the system  $G(s) = \frac{40}{(s+1)(s+10)^2}$

$$\gamma_M = 60^\circ, N_i = 3, \alpha = 0.3$$

1. What is the phase shift from the I-part at the crossover frequency?

$$\varphi_i = -\arctan\left(\frac{1}{N_i}\right) = -\arctan\left(\frac{1}{3}\right) = -0.3218 \text{ rad} = -18.4349^\circ$$

2. What is the maximum phase shift from the Lead-part?

$$\varphi_m = \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) = \arcsin\left(\frac{0.7}{1.3}\right) = 0.5686 \text{ rad} = 32.579^\circ$$

3. What is the phase of  $G(s)$  at the frequency where we want to have the given phase margin (i.e. at the new crossover frequency of the open-loop transfer function)?

$$\angle G(j\omega_c) = \gamma_M - \varphi_i - 180^\circ - \varphi_m = 60^\circ + 18.4349^\circ - 180^\circ - 32.579^\circ = -134.144^\circ$$

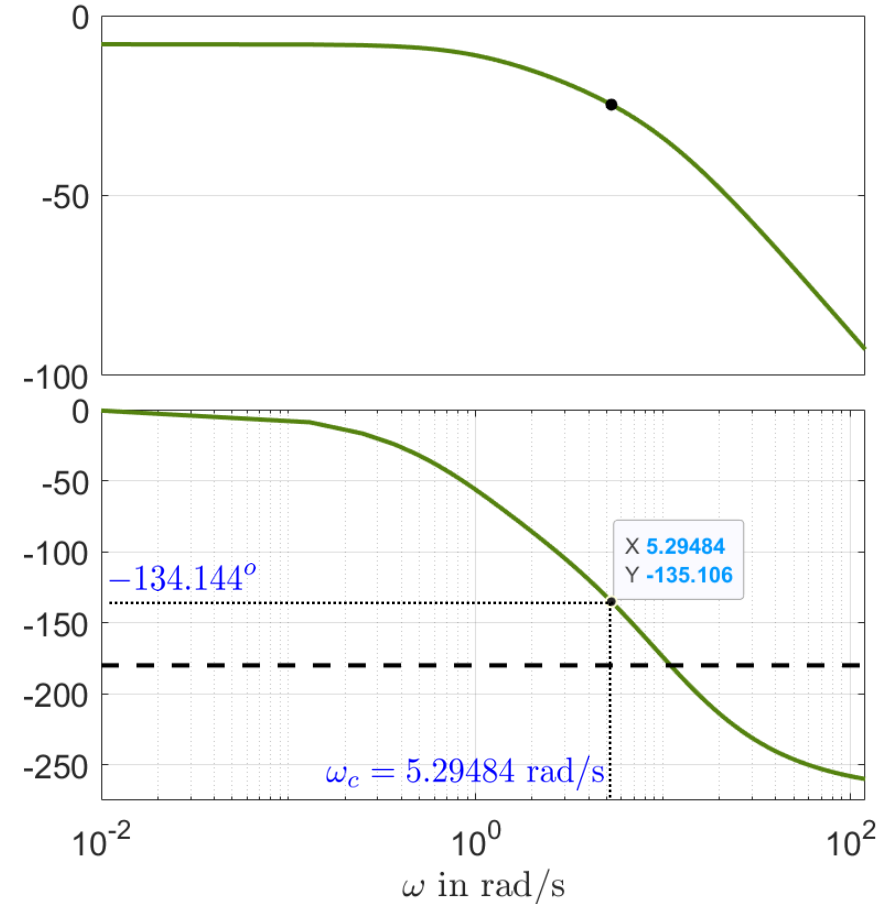
# PI-Lead design – Proportional-Integral-“Differential”

Example: Design a PI-Lead controller for the system

$$G(s) = \frac{40}{(s+1)(s+10)^2} \quad \gamma_M = 60^\circ, N_i = 3, \alpha = 0.3$$

1. What is the phase shift from the I-part at the crossover frequency?  $\varphi_i = -18.4349^\circ$
2. What is the maximum phase shift from the Lead-part?  
 $\varphi_m = 32.579^\circ$
3. What is the phase of  $G(s)$  at the frequency where we want to have the given phase margin (i.e. at the new crossover frequency of the open-loop transfer function)?  
 $\angle G(j\omega_c) = -134.144^\circ$
4. What is the new crossover frequency of the open-loop transfer function?

From the bode plot we get  $\omega_c = 5.29484$  rad/s



# PI-Lead design – Proportional-Integral-“Differential”

Lecture\_08\_exercise\_3\_solution.m

Example: Design a PI-Lead controller for the system

$$G(s) = \frac{40}{(s+1)(s+10)^2} \quad \gamma_M = 60^\circ, N_i = 3, \alpha = 0.3$$

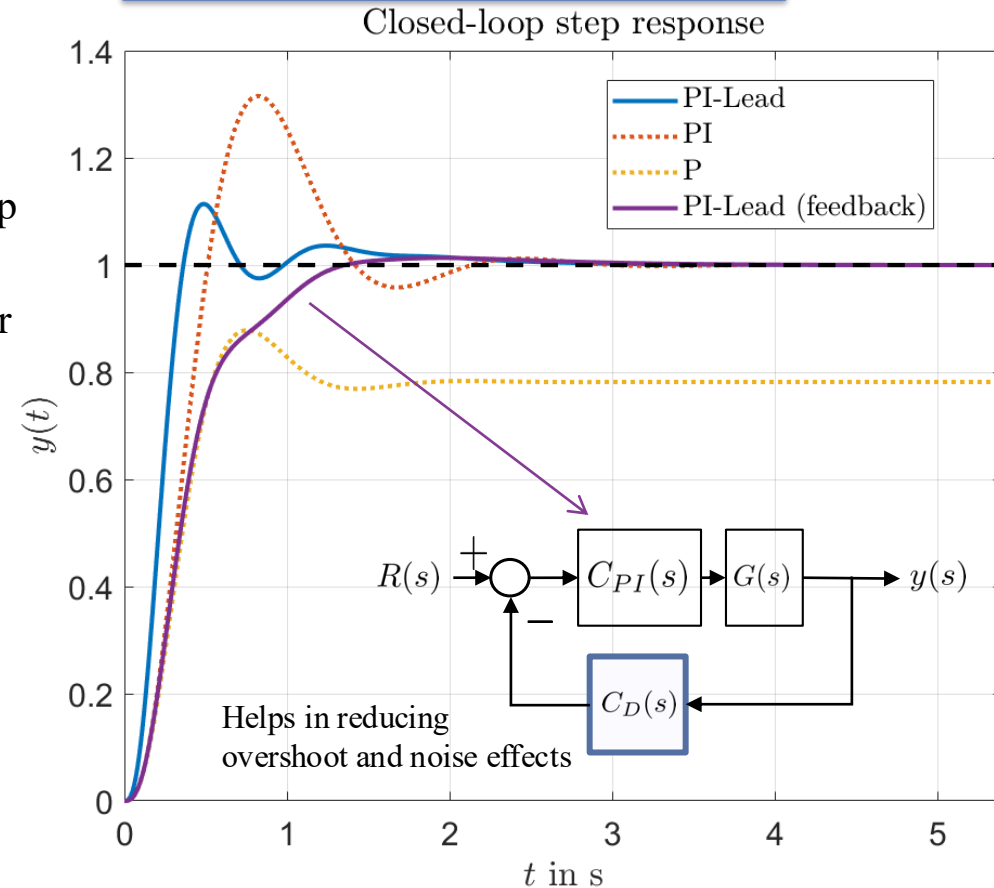
4. What is the new crossover frequency of the open-loop transfer function?  $\omega_c = 5.29484$  rad/s
5. What should the PI-Lead parameters  $\tau_i, \tau_d, K_P$  be for this control objective?

$$\tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.3448$$

$$\tau_i = N_i \frac{1}{\omega_c} = 0.5666$$

$$K_P = \frac{1}{|C_{PI}(s)C_D(s)G(s)|} \Big|_{s=j\omega_c} = 8.9622$$

6. Plot the response of the closed-loop system to a step input.



# PI-Lead design – Proportional-Integral-“Differential”

Design a PI-Lead controller for a given system  $G(s)$ : [Cookbook recipe](#)

- **Step 1:** Select  $\gamma_M, N_i, \alpha$  baring in mind the properties of each parameter (see table on next slide)
- **Step 2:** Calculate  $\varphi_i$  and  $\varphi_m$  from (remember to convert to degrees)

$$\varphi_i = -\arctan\left(\frac{1}{N_i}\right), \quad \varphi_m = \arcsin\left(\frac{1-\alpha}{1+\alpha}\right)$$

- **Step 3:** Calculate the phase of  $G(s)$  at the new crossover frequency from the phase balance equation

$$\varphi_G \triangleq \gamma_M - \varphi_i - 180^\circ - \varphi_m$$

- **Step 4:** Find  $\omega_c$  from the bode plot of  $G(s)$  as the frequency at which  $\angle G(j\omega_c) = \varphi_G$ .
- **Step 5:** Find the remaining parameters  $\tau_i, \tau_d, K_P$  from

$$\tau_d = \frac{1}{\omega_c \sqrt{a}}, \quad \tau_i = N_i \frac{1}{\omega_c}, \quad K_P = \frac{1}{|C_{PI}(s)C_D(s)G(s)|} \Big|_{s=j\omega_c}$$

- **Step 6:** Plot the step response of the closed-loop system to and possibly iterate with new  $\gamma_M, N_i, \alpha$ .

# PI-Lead design – Proportional-Integral-“Differential”

Design a PI-Lead controller for a given system  $G(s)$ : [Cookbook recipe](#)

- Practical properties of  $\gamma_M, N_i, \alpha$ .

|            | Large                                                                                                                                                                                                                                           | Small                                                                                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\gamma_M$ | <ul style="list-style-type: none"> <li>✓ Robustness</li> <li>✓ Small overshoot</li> </ul>                                                                                                                                                       | <ul style="list-style-type: none"> <li>✗ Sensitivity to disturbances</li> <li>✗ Large overshoot</li> </ul>                                                                                                                              |
|            | • Good starting point at 60° - must be positive for stability                                                                                                                                                                                   |                                                                                                                                                                                                                                         |
| $N_i$      | <ul style="list-style-type: none"> <li>✓ Small negative phase shift (reduces overshoot)</li> <li>✗ Slow error correction</li> </ul>                                                                                                             | <ul style="list-style-type: none"> <li>✗ Large negative phase shift (increases overshoot)</li> <li>✓ Fast error correction</li> </ul>                                                                                                   |
|            | • Good starting point at 3 – typically in the range of [2,10]                                                                                                                                                                                   |                                                                                                                                                                                                                                         |
| $\alpha$   | <ul style="list-style-type: none"> <li>✗ Small positive phase shift (smaller phase margin → increased overshoot)</li> <li>✗ Large settling time (slow response)</li> <li>✓ Smaller amplification of noise and high-frequency content</li> </ul> | <ul style="list-style-type: none"> <li>✓ Large positive phase shift (greater phase margin → reduces overshoot)</li> <li>✓ Small settling time (fast response)</li> <li>✗ High gain at high frequencies (noise amplification)</li> </ul> |
|            | • Good starting point at 0.2 – typically in the range of [0.02,1]                                                                                                                                                                               |                                                                                                                                                                                                                                         |

# Today you learned

- What a PI-Lead controller is and which parts it consists of.
- How to design a P controller based on phase margin specifications.
- How to design a PI controller based on phase margin specifications.
- How to design a PI-Lead controller based on phase margin specifications.

# Next lecture 9

Design based on time-domain specifications

- Transform closed-loop specification into open-loop specifications for design.
- Decide which controller parts are needed for a given system.
- Design feedback controllers based on closed-loop system time-domain specifications.

## Today's Exercises

Group work: Position control of the REGBOT